

TIANREN CHEN

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Davis, California

SUMMARY

M.S. student in Computer Science with a strong undergraduate focused on Machine Learning and Deep Learning, with hands-on experience building ML pipelines, recurrent neural networks, and graph neural networks. Strong foundation in Python, backend automation, and data processing.

EDUCATION

University of California, Davis **Sept 2026 – June 2028**
Master of Science in Computer Science (Focus: Machine Learning and Deep Learning) *Davis, California*

University of California, Davis **Sept 2023 – June 2025**
Bachelor of Science in Computer Science *Davis, California*

PROJECTS

Pre-Release IMDb Rating Prediction Python, Pandas, scikit-learn, TensorFlow/Keras

- Co-developed an end-to-end regression pipeline to predict pre-release IMDb ratings by integrating IMDb, TMDb, and social-media metadata and processing 53,005 movie records.
- Engineered actor, director, writer, genre, budget, and popularity features; cleaned missing and inconsistent records to produce a modeling dataset of 42,766 movies.
- Trained and tuned Linear Regression, Random Forest, and neural network models using cross-validation and grid search, achieving 0.909 R^2 and 0.480 RMSE with Random Forest.

Deep Learning for Sequential and Graph-Structured Data Python, PyTorch, scikit-learn

- Implemented RNN, LSTM, and GRU architectures for binary text classification using word embeddings, multilayer recurrent networks, dropout, and early stopping.
- Conducted 18 ablation experiments across embedding size, hidden dimensions, network depth, dropout, learning rate, weight decay, and batch size, achieving up to 86.5% accuracy and 86.7% F1-score.
- Reproduced a two-layer GCN from Kipf and Welling's ICLR 2017 paper and performed dataset-specific ablation studies on Cora, Citeseer, and PubMed, achieving up to 82.4% test accuracy on Cora.

WORK EXPERIENCE

Megawatt, Data Center Network Deployment Intern June 2026 – August 2026

- Validated and troubleshoot server-to-leaf and leaf-to-spine connectivity by analyzing port status, endpoint mappings, Tx/Rx polarity, link speed, and physical-layer failures.
- Diagnosed fiber and InfiniBand link failures using BER testing, optical diagnostics, cable tracing, and port verification in large-scale data center environments.

American Eagle Immigration Law Firm, Software Developer July 2025 – December 2025

- Developed a Python/Flask automation system to extract, transform, validate, and generate structured immigration application data.
- Built backend data-processing workflows with schema validation, normalization, exception handling, and missing-data detection.
- Automated document-generation workflows by integrating structured client data with application logic, reducing manual data-entry requirements.

Golden Fastrack Corp., Summer Intern June 2023 – September 2023

- Wrote SQL queries and performed data-integrity checks to maintain business records and identify duplicate, missing, and inconsistent data.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL

ML & AI: PyTorch, TensorFlow/Keras, Scikit-learn, Machine Learning, Deep Learning, Neural Networks, Model Evaluation, Feature Engineering

Data: NumPy, Pandas, MySQL, SQLite

Software & Infrastructure: Flask, REST APIs, Docker, Git/GitHub, Linux